

Christopher Musco

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Academic Positions

New York University <i>Assistant Professor, Tandon School of Engineering</i> <i>Department of Computer Science and Engineering</i>	New York, NY <i>2019 - present</i>
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Microsoft Research <i>Visiting Researcher, Machine Learning and Optimization Group</i>	Redmond, WA <i>Summer 2020</i>
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Princeton University <i>Research Instructor, Department of Computer Science</i>	Princeton, NJ <i>2018 – 2019</i>
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Massachusetts Institute of Technology <i>Research and Teaching Assistant, Theory of Computation Group</i>	Cambridge, MA <i>2013 – 2018</i>
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Yahoo Labs <i>Research Intern, Scalable Machine Learning Group</i>	New York, NY <i>Summer 2015</i>
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Research Areas

theory of algorithms • randomized methods • numerical linear algebra • sketching and streaming • foundations of data science • network science • machine learning • active learning and function approximation

Education

Massachusetts Institute of Technology <i>Ph.D. in Computer Science</i>	Cambridge, MA <i>2013 – 2018</i>
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Advisor: Jonathan Kelner

Ph.D. Thesis: Faster Linear Algebra for Data Analysis and Machine Learning

M.S. Thesis: Dimensionality Reduction for Sparse and Structured Matrices

Yale University <i>B.S. in Applied Mathematics, B.S. in Computer Science</i>	New Haven, CT <i>2008 – 2012</i>
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Publications

*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science. Any exceptions are marked with a *, which indicates that authors are ordered by contribution.*

Spatially-Aware Leverage Score Sampling via the Pivotal Method. Xiaoou Cheng, Christopher Musco, Atsushi Shimizu, Jonathan Weare. *Preprint* 2023.

Simple Analysis of Priority Sampling. Majid Daliri, Juliana Freire, Christopher Musco, Aécio Santos, Haoxiang Zhang. *Preprint* 2023.

Efficient Block Approximate Matrix Multiplication*. Chuhan Yang, Christopher Musco. *European Symposium on Algorithms (ESA)* 2023.

Moments, Random Walks, and Limits for Spectrum Approximation. Yujia Jin, Christopher Musco, Apoorv Singh, Aaron Sidford. *Conference on Learning Theory (COLT)* 2023.

On the Unreasonable Effectiveness of Single Vector Krylov Methods for Low-Rank Approximation. Raphael Meyer, Cameron Musco, Christopher Musco. *Preprint* 2023.

Dimensionality Reduction for General KDE Mode Finding. Xinyu Luo, Christopher Musco, Cas Widdershoven. *International Conference on Machine Learning (ICML)* 2023.

Near-Optimality Guarantees for Approximating Rational Matrix Functions by the Lanczos Method. Noah Amsel, Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *Preprint* 2023.

Weighted Minwise Hashing Beats Linear Sketching for Inner Product Estimation. Aline Bessa, Majid Daliri, Juliana Freire, Cameron Musco, Christopher Musco, Aécio Santos, Haoxiang Zhang. *ACM Symposium on Principles of Database Systems (PODS)* 2023.

Low-memory Krylov subspace methods for optimal rational matrix function approximation. Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *SIAM Journal on Matrix Analysis and Applications (SIMAX)* 2023.

Active Learning for Single Neuron Models with Lipschitz Non-Linearities. Aarshvi Gajjar, Chinmay Hegde, Christopher Musco. *International Conference on Artificial Intelligence and Statistics (AISTATS)* 2023. Preliminary version at NeurIPS 2022 Workshop on The Symbiosis of Deep Learning and Differential Equations II (DLDE 2022).

Near-Linear Sample Complexity for L_p Polynomial Regression. Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff, Samson Zhou. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2023.

A Tight Analysis of Hutchinson's Diagonal Estimator. Prathamesh Dharangutte, Christopher Musco. *SIAM Symposium on Simplicity in Algorithms (SOSA)* 2023.

Active Linear Regression for ℓ_p Norms and Beyond. Cameron Musco, Christopher Musco, David P. Woodruff, Taisuke Yasuda. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2022.

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How to Quantify Polarization in Models of Opinion Dynamics. Christopher Musco, Indu Ramesh, Johan Ugander, R. Teal Witter. *International Workshop on Mining and Learning with Graphs (at KDD)* 2022.

Chemically-informed data-driven optimization (ChDDO): Leveraging physical models and Bayesian learning to accelerate chemical research*. Daniel Frey, Juhee Shin, Christopher Musco, Miguel A. Modestino. *Reaction Chemistry & Engineering* 2022.

Streaming Approach to In Situ Selection of Key Time Steps for Time-Varying Volume Data*. Mengxi Wu, Yi-Jen Chiang, Christopher Musco. *EuroVis* 2022.

Sublinear Time Spectral Density Estimation. Vladimir Braverman, Aditya Krishnan, Christopher Musco. *ACM Symposium on Theory of Computing (STOC)* 2022.

A Sketch-based Index for Correlated Dataset Search*. Aécio Santos, Aline Bessa, Christopher Musco, Juliana Freire. *IEEE 38th International Conference on Data Engineering (ICDE)* 2022.

Fast Regression for Structured Inputs. Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff Sidford, Samson Zhou. *International Conference on Learning Representations (ICLR)* 2022.

Active Linear Regression in the Online Setting via Leverage Score Sampling. Harvineet Singh, Christopher Musco, Rumi Chunara. *ICML Workshop on Adaptive Experimental Design and Active Learning in the Real World* 2022.

Error Bounds for Lanczos-based Matrix Function Approximation. Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *SIAM Journal on Matrix Analysis and Applications (SIMAX)* 2022.

Fast and Near-Optimal Diagonal Preconditioning. Arun Jambulapati, Jerry Li, Christopher Musco, Aaron Sidford, Kevin Tian. *Preprint* 2021.

Dynamic Trace Estimation. Prathamesh Dharangutte, Christopher Musco. *Conference on Neural Information*

Processing Systems (NeurIPS) 2021.

Finding an Approximate Mode of a Kernel Density Estimate. Jasper C.H. Lee, Jerry Li, Christopher Musco, Jeff M. Phillips, Wai Ming Tai. *European Symposium on Algorithms (ESA)* 2021.

Correlation Sketches for Approximate Join-Correlation Queries*. Aécio Santos, Aline Bessa, Fernando Chirigati, Christopher Musco, Juliana Freire. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

Public Transport Planning: When Transit Network Connectivity Meets Commuting Demand*. Sheng Wang, Yuan Sun, Christopher Musco, Zhifeng Bao. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

Simple Heuristics Yield Provable Algorithms for Masked Low Rank Approximation. Cameron Musco, Christopher Musco, David Woodruff. *Innovations in Theoretical Computer Science (ITCS)* 2021.

Hutch++: Optimal Stochastic Trace Estimation. Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff. *SIAM Symposium on Simplicity in Algorithms (SOSA)* 2021.

Graph Learning for Inverse Landscape Genetics. Prathamesh Dharangutte, Christopher Musco. *AAAI Conference on Artificial Intelligence (AAAI)* 2021. Short version in NeurIPS 2020 AI for Earth Sciences workshop.

Fourier Sparse Leverage Scores and Approximate Kernel Learning. Tamás Erdélyi, Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020. Invited for spotlight presentation (4% of submissions).

Near Optimal Linear Algebra in the Online and Sliding Window Models. Vladimir Braverman, Petros Drineas, Cameron Musco, Christopher Musco, Jalaj Upadhyay, David P. Woodruff, Samson Zhou. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2020.

The Statistical Cost of Robust Kernel Hyperparameter Tuning. Raphael Meyer, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020.

Low-Rank Toeplitz Matrix Estimation via Random Ultra-Sparse Rulers. Hannah Lawrence, Jerry Li, Cameron Musco, Christopher Musco. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2020.

Sample Efficient Toeplitz Covariance Estimation. Yonina Eldar, Jerry Li, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Fast and Space Efficient Spectral Sparsification in Dynamic Streams. Michael Kapralov, Aida Mousavifar, Cameron Musco, Christopher Musco, Navid Nouri, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Analyzing the Impact of Filter Bubbles on Social Network Polarization. Uthsav Chitra, Christopher Musco. *ACM International Conference on Web Search and Data Mining (WSDM)* 2020. Preliminary version in KDD Workshop on Issues of Sentiment Discovery and Opinion Mining.

Projection-Cost-Preserving Sketches: Proof Strategies and Constructions . Cameron Musco, Christopher Musco. *Technical Report* 2020.

A Universal Sampling Method for Reconstructing Signals with Simple Fourier Transforms. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ACM Symposium on Theory of Computing (STOC)* 2019.

Inferring networks from random walk-based node similarities. Jeremy Hoskins, Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *Conference on Neural Information Processing Systems (NeurIPS)* 2018.

Eigenvector Computation and Community Detection in Asynchronous Gossip Models. Frederik Mallmann-Trenn, Cameron Musco, Christopher Musco. *International Colloquium on Automata, Languages, and Programming (ICALP)* 2018.

Minimizing Polarization and Disagreement in Social Networks. Cameron Musco, Christopher Musco,

Charalampos Tsourakakis. *The Web Conference (WWW)* 2018.

Stability of the Lanczos Method for Matrix Function Approximation. Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2018.

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2017.

Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML)* 2017.

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX)* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML)* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2015. Invited for full oral presentation (1 of 15, < 1% of submissions). Also presented at the 2016 Copper Mountain Conference on Iterative Methods.

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of Computing (STOC)* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiroglou-Douskos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS)*. 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS)* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2014, Special Issue SIAM Journal on Computing 2017.

Funding

NSF CAREER Award

Frontiers in Matrix Sketching.

2021 - 2026

PI, \$562,224

NSF III Medium

Dataset Search and Ranking for Data Augmentation and Explanation.

2021 - 2024

Co-PI (w/ Juliana Freire, NYU), \$1,093,195

DOE Office of Science Grant

Leverage Score Sampling for Parametric PDEs.

2021 - 2023

PI (w/ Chinmay Hegde, NYU), \$300,000

NSF/CRA

CIFellow Fellow Award for Dr. Aline Bessa.

2021-2022

PI, \$263,250

Current Graduate Student and Postdoc Advising

Tyler Chen. Postdoc, New York University.

2022 - present

Noah Amsel. Ph.D., New York University (with Prof. Joan Bruna).	2022 - present
Majid Daliri. Ph.D., New York University.	2022 - present
Aarshvi Gajjar. Ph.D., New York University (with Prof. Chinmay Hedge).	2021 - present
Apoorv Singh. Ph.D., New York University.	2020 - present
R. Teal Witter. Ph.D., New York University (with Prof. Lisa Hellerstein).	2020 - present
Chuhan Yang. Ph.D., New York University (primary advisor is Prof. Saif Jabari).	2020 - present
Raphael Meyer. Ph.D., New York University.	2019 - present

Past Advising

Aline Bessa. Postdoc, New York University.	2021
David Persson. Visiting Ph.D. student.	2023
Siddharth Sagar. M.S., New York University.	2023
Yajur Ahuja. M.S., New York University.	2023
Sundeeep Chenreddy M.S., New York University.	2023
Atsushu Shimuzu. M.S., New York University.	2022 - 2023
Danrong Li. M.S., New York University.	2022 - 2023
Indu Ramesh. M.S., New York University.	2020 - 2022
Xinyu Luo. M.S., New York University.	2020 - 2022
Prathamesh Dharangutte. M.S., New York University.	2019 - 2021
John Shin. M.S., New York University.	2019 - 2020

I have advised over a dozen undergraduate guided studies and research projects, as well as several research projects involving high school students.

Ph.D. Thesis Committees

Ishan Agarwal , NYU	Defended 2023.
Yiming Zhang , NYU	Defended 2022.
Francis Williams, NYU	Defended 2021.

Teaching

Machine Learning, Graduate (NYU CS-GY 6923) <i>Course Instructor</i>	Spring 2022, 2023
Algorithmic Machine Learning and Data Science (NYU CS-GY 6763) <i>Course Instructor</i>	Fall 2019 - 2023
Introduction to Machine Learning (NYU CS-UY 4563) <i>Course Instructor</i>	Spring 2020
Advanced Algorithm Design (Princeton COS 521) <i>Course Instructor</i>	Fall 2018
Advanced Algorithms (MIT 6.854/18.415) <i>Teaching Assistant</i>	Spring 2016

Technical Communication Skills for Graduate Students (MIT 6.S977)

Workshop Leader

Spring 2016

Mentorship and Outreach

SIAM/MathWorks Math Modeling Challenge

Problem Development Committee, Director of Technical Computing Judging

2018 - present

Support SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Serve on committee in charge of writing practice and contest problems. Lead judging and award presentation for student Technical Computing prizes.

MIT Graduate Communication Lab

Communication Adviser

2015 - 2018

Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offered feedback, guidance, and coaching to individual students during weekly office hours.

Research Science Institute at MIT

Summer Research Adviser

2014

Primary supervisor for a high school student participating in the prestigious RSI summer research program. Advisee's project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Invited Talks

On the Effectiveness of Single Vector Krylov Methods for Low-rank Approximation

Foundations of Computational Mathematics (FoCM)

June 2023

Tutorial on Matrix Sketching

BIRS Workshop on Perspectives on Matrix Computations

Mar. 2023

Integrating Education and Research in Early Career Funding Proposals

Joint Mathematics Meetings, SIAM Special Session on Education

Jan. 2023

Robust Active Learning via Leverage Score Sampling

Canadian Mathematics Society, Winter Meeting

Dec. 2022

Iterative Sampling Methods for Matrix Approximation

University of Michigan, Randomized Numerical Linear Algebra Seminar

Oct. 2022

Noisy Implicit Methods for Matrix Functions

SIAM Annual Meeting

July 2022

Sublinear Time Spectral Density Estimation

Queens College CUNY Computer Science Colloquium

Mar. 2023

National University of Singapore, Workshop on Algorithms and Foundations for Data Science

June 2022

Oxford University, Computational Mathematics and Applications Seminar

Feb. 2022

University of British Columbia, Applied Math Seminar

Jan. 2022

Workshop on Algorithms for Large Data

Aug. 2021

Cornell University, Scientific Computing and Numerics Seminar

Mar. 2021

Optimal Stochastic Trace Estimation

New York University, Numerical Analysis and Scientific Computing Seminar

Apr. 2022

University of Michigan, Randomized Numerical Linear Algebra Seminar

Nov. 2021

Stanford University, Operations Research Seminar

Oct. 2020

A TCS Perspective on Iterative Eigensolvers

Complexity of Matrix Computations Seminar	Oct. 2021
Mathematical Modeling and Coding in the Classroom	
Mu Alpha Theta National Convention	July 2021
Randomized Algorithms for Probing the Spectrum of a Matrix	
George Mason University, AI Tea Series	Apr. 2021
Importance Sampling for Approximating High-Dimensional Kernel Matrices	
Emory University, Numerical Analysis and Scientific Computing Seminar	Feb. 2021
Structured Covariance Estimation	
Oxford Statistics and Machine Learning in Finance Seminar	June 2022
Yahoo! Research, Seminar	Apr. 2020
Johns Hopkins University, Theory Seminar	Oct. 2019
From Randomized Linear Algebra to Randomized Functional Analysis	
SIAM Mathematics of Data Science Conference	May 2020
University of Utah, Data Science Seminar	Jan. 2020
Rutgers University, DIMACS RandNLA Workshop	Sept. 2019
Learning Signals with Simple Fourier transforms	
Columbia University, Theory Seminar	Oct. 2018
UC Berkeley, Simons Institute Workshop	Sept. 2018
Princeton University, Theory Seminar	Sept. 2018
The Lanczos Method in Data Science: Challenges and Opportunities	
SIAM Conference on Parallel Processing for Scientific Computing, Minisymposium	Feb. 2020
Georgetown University, CS Seminar	Feb. 2018
Unifying Random Features and Leverage Scores for Kernel Matrix Approximation	
Institute for Advanced Study, Computer Science and Discrete Mathematics Seminar	Feb. 2018
Harvard University, guest lecture in "Sketching Algorithms for Big Data"	Nov. 2017
Microsoft Research, Redmond, Research Seminar	Sept. 2017
Sketching and Sampling Methods for Large Scale Linear Algebra	
Twitter, Cambridge, Research Seminar	July 2017
Recursive Sampling for the Nyström Method	
New England Machine Learning Day	May 2017
Introduction to Linear Sketching	
MIT, Theory Retreat	Sept. 2016
Ridge Leverage Score Sampling	
National Institute of Informatics, Shonan Meeting	July 2016
MIT, Algorithms & Complexity Seminar	July 2016
University of Utah, Data Group Meeting	Jan. 2016
Large Scale Column Subset Selection	
Yahoo Labs, Science Week	July 2015
Dimensionality Reduction for k-Means Clustering	
IBM T.J. Watson, Mathematical Sciences Research Seminar	Aug. 2015
MIT, Theory Lunch	June 2014

Uniform Sampling for Matrix Approximation

Yahoo Labs, Scalable Machine Learning Research Seminar

June 2015

Single Pass Spectral Sparsification in Dynamic Streams

MIT, Annual Sublinear Algorithms Day

Apr. 2015

Harvard University, Theory Seminar

Nov. 2014

Service

Organization.....

NYU Theoretical Computer Science Seminar

2022 - present

NYU Computer Science Theory "Pandemic Presentations" Workshop

2022

SIAM Annual Meeting Minisymposium Organizer

2021

SIAM Mathematics of Data Science Minisymposium Organizer

2020

Department Service.....

Faculty Search Committee

2019, 2021, 2022

Chair, Ph.D. Admissions Committee

2021 - present

Department Chair Search Committee

2021 - 2023

Online Faculty Oversight Committee

2021 - 2023

Conference Programming.....

STOC Program Committee Member

2023

FOCS Program Committee Member

2022

ICML Area Chair

2023

NeurIPS Area Chair

2019, 2021, 2022, 2023

AAAI Senior Area Chair

2022

Journal Editing.....

Editor, SICOMP Special Issue for STOC

2023

Conference Reviewing.....

STOC, FOCS, SODA, ICALP, ITCS, ICML, NeurIPS, COLT, AAAI, AISTATS, SOCG, ESA, APPROX, RANDOM, STACS, IWOCA, WALCOM, NDSS

Journal Reviewing.....

Journal of the ACM, SIAM Journal on Computing, ACM Transactions on Algorithms, Theoretical Computer Science, Journal of Machine Learning Research, SIAM Journal on Matrix Analysis and Applications, SIAM Journal on Scientific Computing, Linear Algebra and Its Applications, SIAM Journal on Mathematics of Data Science, Science Advances, Mathematical Programming, IEEE Transactions on Signal Processing, IEEE Transactions on Knowledge and Data Engineering, International Journal of Foundations of Computer Science, IEEE Journal on Selected Areas in Information Theory, BIT Numerical Mathematics, AMS Notices

Grants Panels.....

National Science Foundation Panel

2019, 2022

NSF Graduate Research Fellowship Reviewer

2022

Department of Energy External Reviewer

2022, 2023

Notables.....

ICML 2018 Outstanding Reviewer, ICML Expert Review 2021

Invited Workshops and Events

Banff International Research Station (BIRS)

Invited participant

2023

Workshop on Perspectives on Matrix Computations: Theoretical Computer Science Meets Numerical Analysis.

Rutgers, DIMACS

Invited participant

2019

Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization.

UC Berkeley, Simons Institute for the Theory of Computing

Invited participant

2018

Workshop on Randomized Numerical Linear Algebra and Applications.

Japan National Institute of Informatics

Invited participant

2016

Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra.

Honors and Awards

NYU Tandon

2023

Junior Faculty Research Award

NYU Tandon

2023

Goddard Junior Faculty Fellowship Award.

NYU Tandon

2022

Nominated for Jacobs Excellence in Education Innovation Award.

National Science Foundation

2014-2017

Graduate Research Fellowship, recipient.

Yale University

2012

Cum Laude, Distinction in Both Majors.

Other Professional Experience

Redfin

Seattle, WA

Software Engineer, Data Team

2012 – 2014

Elysium Digital

Cambridge, MA

Summer Consultant

2011

Amicus

New Haven, CT

Software Developer

2010 – 2011